

Individual Reflection

Within this reflective piece I will look back at key moments in this module where I employed a degree of self-reflection and determination to achieve the learning outcomes. These outcomes are listed, where relevant, underneath the examples of work in which they are demonstrated. To structure my insights, I have employed Rolfe, Freshwater & Jasper's (2001) 'What?', 'So what?', 'What next?' format.

What?

Despite a lack of experience in the field of information security, I was determined to keep previous reflections in mind when beginning this module, remembering that I needed to trust that I can quickly acquire knowledge and understanding:

I have come to realise that much of the knowledge gained from this module can be directly used in my role as an IT Technician: the research into digital transformation and subsequent security adoption can inform strategies to help me to better facilitate the uptake of more secure digital practices. Furthermore, I have very recently been asked to develop a new website for my school and so my new-found abilities to conduct and interpret vulnerability analyses has become an important skill to utilise.

In terms of my future studies, I have learnt to have more belief in my ability to quickly build on knowledge, and I now understand better that knowledge breeds motivation (Shapiro et al., 2017). Ultimately, I have learnt that I should aim to keep an open mind about new modules, not writing them off as being 'not my area' and not writing myself off if I feel that I initially have less knowledge than others.

Figure 1: Reflections from Network Security Module (Edge, 2025).

I was, however, somewhat disappointed to receive feedback from my first assessment that suggested that my report had lacked depth:

Overall, the report effectively addresses the key elements of risk identification but would benefit from stronger justification of methodology, more specific application, deeper critical analysis, and improved clarity in structure and presentation. Enhancing the depth of evaluation through comparative analysis, empirical examples, and structured risk prioritization will elevate the quality of the submission.

Figure 2: Excerpt from tutor feedback for Assessment 1 (Tahir, 2025).

When replying to a peer within the collaborative discussion in Units 1 and 2, I further reflected on a lack of depth in my work when comparing it to a peer's impressive contribution:



In reply to Timothy Brayshaw

Peer response

by [Todd Edge](#) - Friday, 18 April 2025, 8:03 AM

Hi Timothy,

You succinctly summarized Kovaitė and Stankevičienė (2019) and Agustian et al.'s (2023) findings and certainly offer more depth here than I did. I think you did very well to find an article that so closely agreed with Kovaitė and Stankevičienė's (2019) findings.

Although I agree that there is crossover between the two articles in terms of the 4 areas of risk you outlined - Technical Risks, Competence Risks, Data Privacy & Security Risks, and Acceptance Risks - Kovaitė and Stankevičienė (2019) focus a lot more on customers as being a key area of risk to business models, with their potential negative behaviours in response to new processes being particularly impactful. Agustian et al (2023), on the other hand seem to only briefly touch upon customer influence in terms of the need to meet their preferences through data analytics and subsequent personalisation of experiences; Kovaitė and Stankevičienė (2019) would perhaps view this as too simplistic a view, not taking into account the fact that all customers cannot be equally pleased, and that there will always be a risk that pockets of customers pose to the business in terms of their rejection of new processes and their keenness to share their displeasure online.

Overall, I enjoyed reading your analysis; it made me realise that I had not added sufficient depth to my original post in regard to the main impacts that digital transformations have on business models and it made me delve deeper into these areas before completing my risk identification report.

References

Agustian, K., Mubarak, E. S., Zen, A., Wiwin and Malik, A. J., (2023) *The impact of digital transformation on business models and competitive advantage. Technology and Society Perspectives (TACIT)*, 1(2), pp. 79–93. Available at: <https://doi.org/10.61100/tacit.v1i2.55>

-

Kovaitė, K. and Stankevičienė, J., (2019) *Risks of digitalisation of business models. Proceedings of the International Scientific Conference "Contemporary Issues in Business, Management and Economics Engineering 2019"*, Vilnius, Lithuania, 9–10 May. Vilnius Gediminas Technical University Press. Available at: <https://doi.org/10.3846/cibmee.2019.039>

Figure 3: My 'Peer Response' for Collaborative Discussion 1

Outcomes achieved:

Identify and analyse security risks, threats and vulnerabilities in information systems and determine appropriate methodologies, tools, and techniques to manage and/or solve them.

Gather and synthesise information from multiple sources (including internet security alerts & warning sites) to aid in the systematic analysis of security breaches & issues.

Critically appraise and utilise methodologies, tools and techniques that help manage and audit security issues

Additionally, a former student's GitHub repository appeared in one of my internet searches when researching third party libraries to aid in the visualisation and management of attack trees. I read his readme file for his assessment 2 submission, which detailed a very impressive graphical front end utilised within his solution, together with screenshots of some beautifully rendered attack trees outputted as image files. As a result, I found myself further determined to produce a comprehensive solution to meet the requirements of assessment 2. This increase in motivation is supported by Ficano's (2012) findings that male students tend to increase their levels of academic performance when exposed to higher-achieving male peers. Xu et al.'s (2023) findings further support this, indicating that positive peer pressure, where there is a high level of peer achievement, results in higher levels of academic motivation.

So what?

I sought to provide a comprehensive solution to assessment 2 that matched those submitted by my peers, while I also saw opportunities to improve in key areas identified by my tutor, seeing that these aligned with some of the assessment 2 guidelines (see Figure 4).

decision to not only provide a visualisation within the terminal but to also add the option to output the attack tree to an image file.

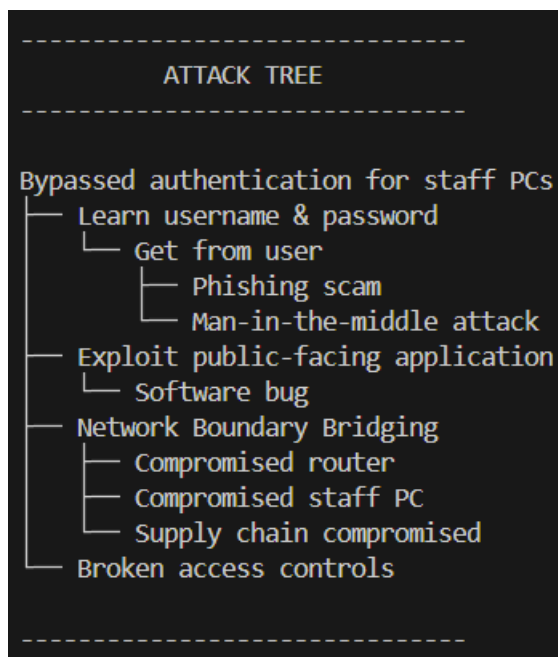


Figure 5: A typical example of an attack tree visualisation printed to the terminal.

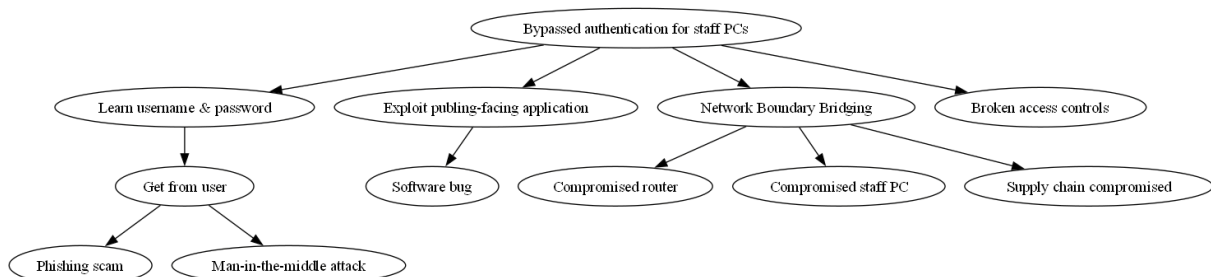


Figure 6: Image file data visualisation

Outcomes achieved:

Gather and synthesise information from multiple sources (including internet security alerts & warning sites) to aid in the systematic analysis of security breaches & issues.

Articulate the legal, social, ethical, and professional issues faced by information security professionals.

In order to meet the requirement of providing a function to '...add values to the leaf nodes' and '...aggregate these values into an overall threat assessment rating...', I not only ensured that these were printed to the terminal but also included the option

for threat probability values and the overall threat assessment rating to be outputted to a text file, with each value carefully formatted for readability:

```
Leaf node 'Sending payload through data packets' has no threat probability value assigned
Leaf node 'Reputable third party developer software updates compromised' has threat probability value: 2.9
Leaf node 'Disreputable third party developer purposefully deploying update with malicious code' has no threat probability value assigned
Leaf node 'Malicious website link clicked on' has no threat probability value assigned
Leaf node 'Masquerading software downloaded' has no threat probability value assigned
Leaf node 'Sending payload via third party smartphone applications' has threat probability value: 4.8

OVERALL THREAT ASSESSMENT RATING: 7.7
```

Figure 7: Threat assessment values outputted in terminal.

```
THREAT PROBABILITY VALUES: Pre-digitalisation AT 3
-----

LEAF NODE:                                     THREAT PROBABILITY VALUE: 0
Sending payload through data packets

LEAF NODE:                                     THREAT PROBABILITY VALUE: 0
Reputable third party developer software updates compromised

LEAF NODE:                                     THREAT PROBABILITY VALUE: 0
Disreputable third party developer purposefully deploying update with malicious code

LEAF NODE:                                     THREAT PROBABILITY VALUE: 0
Malicious website link clicked on

LEAF NODE:                                     THREAT PROBABILITY VALUE: 2.3
Masquerading software downloaded

LEAF NODE:                                     THREAT PROBABILITY VALUE: 0
Sending payload via third party smartphone applications

-----
OVERALL THREAT ASSESSMENT RATING: 2.3
-----
```

Figure 8: Threat assessment values outputted as text file.

Outcomes achieved:

Gather and synthesise information from multiple sources (including internet security alerts & warning sites) to aid in the systematic analysis of security breaches & issues.

To address requirement 3 in Figure 4, I sought to clearly explain coding decisions within comments, to provide a ‘... stronger justification of methodology.’ (Tahir, 2025).

```
DotExporter(root).to_dotfile(f"{file_name}.dot") # Pass the nodes of the tree to the Dotxcporter
# method that outputs the nodes to a dot file that can
# be processed by image_output_confirmed() to output a
# graphical representation of the file, in JPEG format.
```

Figure 9: Comment regarding code decision justification within functions.py

```

## The below processes were chosen instead of the more efficient to_picture
# method (c0fec0de, 2023) as this kept resulting in write permission denial.

# Read dot file of nodes and output as PNG file in inputted path.
try:
    subprocess.check_call([
        "dot",
        f"{file_name}.dot",
        "-Tpng",
        "-o",
        combine_file_paths(directory, file_name) + ".png"
    ])
    return True

```

Figure 10: Comment regarding code decision justification within functions.py

```

PIC_CHOICE = None # Initialise variable so that line 75 can access it
                  # if condition on line 52 is not met.

```

Figure 11: Comment regarding code decision justification within main.py

Improvements in Presentation and Structure

As seen in Figure 5, I gave a good deal of thought about the presentation of the application's outputted information and made changes along the way to ensure that information was displayed more clearly and concisely.

THREAT PROBABILITY VALUES FOR test	
Leaf node: sub0A	Threat probability value: 2.0
Leaf node: sub0B	Threat probability value: 9.9
Leaf node: sub1	Threat probability value: 3.2
TOTAL THREAT PROBABILITY SCORE: 15.1	

THREAT PROBABILITY VALUES: Pre-digitalisation AT 3	
LEAF NODE: Sending payload through data packets	THREAT PROBABILITY VALUE: 0
LEAF NODE: Reputable third party developer software updates compromised	THREAT PROBABILITY VALUE: 0
LEAF NODE: Disreputable third party developer purposefully deploying update with malicious code	THREAT PROBABILITY VALUE: 0
LEAF NODE: Malicious website link clicked on	THREAT PROBABILITY VALUE: 0
LEAF NODE: Masquerading software downloaded	THREAT PROBABILITY VALUE: 2.3
LEAF NODE: Sending payload via third party smartphone applications	THREAT PROBABILITY VALUE: 0
OVERALL THREAT ASSESSMENT RATING: 2.3	

Figure 12: Earlier vs later formatting changes to the presentation of the outputted threat values text file.

This careful consideration of presentation extended to information printed to the terminal (see Figure 4 & Figure 13).

```
For which leaf node would you like to add a threat probability value?  
  
OPTIONS: Phishing scam  
         Man-in-the-middle attack  
         Software bug  
         Compromised router  
         Compromised staff PC  
         Supply chain compromised  
         Broken access controls
```

Figure 13: Clearly presented leaf node options printed to the terminal.

Outcomes achieved:

Gather and synthesise information from multiple sources (including internet security alerts & warning sites) to aid in the systematic analysis of security breaches & issues.

To show an improved clarity of structure, I separated my application's functions and command line interface (CLI) logic into 2 distinct modules – functions.py and main.py. Further aiding presentation, comprehensive module docstrings presented each module clearly and concisely.


```

"""
CLI module for building and editing attributes for an attack tree loaded from
a JSON file.

This module loads the JSON file and builds an on-screen visual representation
of the attack tree (allowing for optional output as a PNG file) and allows
users to assign threat probability values to each attack tree node before aggregating these
scores and providing an optional text file output of said scores.

Dependencies:
- json
- anytree (RenderTree) # c0fec0de(2023)
- functions.py (see below for list of functions used from this file)

Functions:
- build_tree(data, parent=None): Constructs a tree from a given dictionary.
- load_json_file(file_name): Loads a JSON file into a dictionary.
- list_leaf_nodes(root): Returns all leaf node names in the tree.
- node_exists(node_name, root): Checks if a leaf node exists in the tree.
- return_matching_node(node_name, root): Retrieves a leaf node by its name.
- convert_to_float(input_value): Converts an input value to a float.
- assign_threat_value(target_node, threat_value, root): Assigns a threat value
  to a leaf node.
- aggregate_threat_scores(root): Aggregates threat values for all leaf nodes.
- print_threat_attributes(root): Displays assigned threat values for leaf nodes.
- export_tree_to_dot_file(root, file_name): Saves the tree structure to a DOT file.
- file_path_is_valid(path): Checks whether inputted file path is valid by verifying
  if the file exists.
- dir_path_is_valid(path): Checks whether inputted directory path is valid by verifying
  if the file exists.
- combine_file_paths(directory, file_name): Combines a directory path with a file name to
  make a full file path.
- output_image(directory, file_name): Processes tree data in dot file and outputs graphical
  PNG file representation of tree.
- record_threat_values_for_all_nodes(root): Records threat values in a
  dictionary list.
- output_threat_values_to_txt_file(list_of_dicts, file_name, root): Exports
  threat values for leaf nodes to a text file.
- choice_confirmed(choice): Confirms whether user input is 'Y' or 'y'.

Usage:
  Run the file as a script and follow on-screen prompts.
"""

```

Figure 14: An example of a module docstring.

Outcomes achieved:

Identify and analyse security risks, threats and vulnerabilities in information systems and determine appropriate methodologies, tools, and techniques to manage and/or solve them.

What next?

I was surprised by how motivated I was to address the areas of improvement set out in my tutor feedback for assessment 1 of this module. Combined with the added

motivation that seemed to be a result of exposure to impressive work contributions from peers, I have realised that seeking out high-performing peers is likely to aid my motivation and performance in future modules. I feel that these realisations will increase my frequency of interactions with both my tutor and peers in my next module.

References

c0fec0de. (2023) Any Python Tree Data. Available from: <https://anytree.readthedocs.io/en/latest/index.html> [Accessed 03 April 2025].

Edge, T. (2025) Individual Reflective Piece. *NS_PCOM7E January 2025*. Assignment submitted to the University of Essex Online.

Ficano, C. C. (2012) Peer effects in college academic outcomes – Gender matters. *Economics of education review* 31 (6): 1102–1115. DOI: <https://doi.org/10.1016/j.econedurev.2012.07.012>

Rolfe, G., Freshwater, D. & Jasper, M. (2001) *Critical reflection in nursing and the helping professions: a user's guide*. Basingstoke: Palgrave Macmillan.

Tahir, S. (2025) Risk Identification Report. *ISM_PCOM7E March 2025*. Tutor Feedback submitted on behalf of University of Essex Online.

The Graphviz Authors. (2025) Graphviz. Available from: <https://graphviz.org/> [Accessed 05 April 2025].

Xu, L., Zhang, J., Ding, Y., Zheng, J., Sun, G., Zhang, W., & Philbin, S. P. (2023) Understanding the role of peer pressure on engineering students' learning behavior: A TPB perspective. *Frontiers in public health* 10: 1069384–1069384. DOI: <https://doi.org/10.3389/fpubh.2022.1069384>